

Technical Document #541: Cybersecurity - Comprehensive Overview

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Security information and event management (SIEM) systems collect and analyze security logs. They provide real-time visibility into security events.

Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Social engineering manipulates people into revealing confidential information. Phishing is the most common social engineering attack.

Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Key Takeaways:

- Multi-factor authentication (MFA) requires multiple forms of verification.
- Network segmentation divides networks into smaller segments.
- Encryption converts plaintext into ciphertext to protect data confidentiality.